

The Strict Derivation of The Dual Tensor Theory of Mass Density: From First Principles to Secondary Energy Emission

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Abstract

This document provides a rigorous mathematical derivation for the “Dual Tensor Theory of Mass Density”. Starting from first principles (the Action Principle and Energy-Momentum Conservation), it establishes total mass density (ρ_m) as the absolute independent variable. Consequently, both the macroscopic curvature of spacetime and the microscopic sectional pressure are derived directly from ρ_m . Furthermore, thermal energy and radiative luminosity are mathematically proven to be strictly secondary byproducts resulting from the critical density threshold (ρ_{crit}).

1 Macroscopic Tensor and Spacetime Curvature via the Action Principle

We begin by formulating the modified Einstein-Hilbert action incorporating the matter density field:

$$\mathcal{S} = \int \left(\frac{c^4}{16\pi G} R + \mathcal{L}_m(\rho_m) \right) \sqrt{-g} d^4x \quad (1)$$

Where R is the Ricci scalar, g is the determinant of the metric tensor $g_{\mu\nu}$, and $\mathcal{L}_m$ is the Lagrangian density of matter dependent fundamentally on the mass density ρ_m .

Taking the variation with respect to the metric tensor $g^{\mu\nu}$ and applying the stationary action principle ($\delta\mathcal{S} = 0$):

$$\frac{\delta\mathcal{S}}{\delta g^{\mu\nu}} = \int \left(\frac{c^4}{16\pi G} \frac{\delta(R\sqrt{-g})}{\delta g^{\mu\nu}} + \frac{\delta(\mathcal{L}_m\sqrt{-g})}{\delta g^{\mu\nu}} \right) d^4x = 0 \quad (2)$$

By defining the stress-energy tensor $T_{\mu\nu}$ as:

$$T_{\mu\nu} \equiv -\frac{2}{\sqrt{-g}} \frac{\delta(\mathcal{L}_m\sqrt{-g})}{\delta g^{\mu\nu}} \quad (3)$$

We yield the macroscopic field equation:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (4)$$

Isolating the primary time-time component as a direct function of mass density ($T_{00} \approx \rho_m c^2$), the macroscopic curvature is explicitly defined:

$$G_{00} = \frac{8\pi G}{c^2} \rho_m \quad (5)$$

This proves that the macroscopic geometric curvature (G_{00}) is a direct, independent differential function of the total density ρ_m .

2 Microscopic Sectional Pressure via Energy Conservation

To prove that internal sectional pressure (P) is a microscopic derivative of density, we apply the covariant conservation of the stress-energy tensor ($\nabla_\mu T^{\mu\nu} = 0$). For a spherically symmetric, static body, this yields the Tolman-Oppenheimer-Volkoff (TOV) equation:

$$\frac{dP}{dr} = -\frac{GM(r)\rho_m(r)}{r^2} \left[1 + \frac{P(r)}{\rho_m(r)c^2} \right] \left[1 + \frac{4\pi r^3 P(r)}{M(r)c^2} \right] \left[1 - \frac{2GM(r)}{c^2 r} \right]^{-1} \quad (6)$$

In the classical limit ($c \rightarrow \infty$), this reduces to the differential hydrostatic equilibrium:

$$\frac{dP}{dr} = -\frac{GM(r)\rho_m(r)}{r^2} \quad (7)$$

Where the enclosed mass is the integral of density over the volume:

$$M(r) = \int_0^r 4\pi r'^2 \rho_m(r') dr' \quad (8)$$

Integrating from the center ($r = 0$) to the surface ($r = R$), assuming a mean density model:

$$P_c = \int_0^{P_c} dP = \int_0^R \frac{GM(r)\rho_m(r)}{r^2} dr \approx C_1 G M^{2/3} \rho_m^{4/3} \quad (9)$$

This confirms that the central sectional pressure (P_c) is an integral derivative arising solely from the cumulative volumetric density (ρ_m).

3 Analytical Derivation of the Critical Density Threshold (ρ_{crit})

For the central pressure P_c to overcome the electromagnetic Coulomb barrier, we define the Coulomb repulsion energy between two nuclei (Z_1, Z_2):

$$E_c = \frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{R_0} \quad (10)$$

The thermal kinetic energy per particle at the core is dictated by the ideal gas equation of state:

$$k_B T_c = \frac{P_c \cdot \mu m_u}{\rho_m} \quad (11)$$

Where μ is the mean molecular weight and m_u is the atomic mass unit. Incorporating the Gamow tunneling factor for quantum penetration, a sustained fusion chain requires the kinetic energy to reach a specific fraction ξ of E_c :

$$k_B T_c \geq \xi E_c \implies \frac{P_c \cdot \mu m_u}{\rho_m} \geq \xi E_c \implies P_c \geq \frac{\xi E_c \rho_m}{\mu m_u} \quad (12)$$

Substituting the derived expression for P_c from Eq. (9):

$$C_1 G M^{2/3} \rho_m^{4/3} \geq \frac{\xi E_c \rho_m}{\mu m_u} \quad (13)$$

Dividing both sides by ρ_m :

$$C_1 G M^{2/3} \rho_m^{1/3} \geq \frac{\xi E_c}{\mu m_u} \quad (14)$$

Cubing both sides yields the exact rigorous condition for the Critical Density Threshold (ρ_{crit}):

$$\rho_{crit} \geq \left(\frac{\xi E_c}{C_1 \mu m_u G M^{2/3}} \right)^3 \quad (15)$$

If $\rho_m < \rho_{crit}$, the micro-tensor fails to trigger fusion.

4 Mass-Energy Liberation Derivative (Secondary Energy)

Once the threshold ($\rho_m \geq \rho_{crit}$) is satisfied, the fusion reaction rate per unit volume (R_{12}) commences:

$$R_{12} = n_1 n_2 \langle \sigma v \rangle \propto \rho_m^2 \cdot T_c^n \quad (16)$$

The time derivative of the liberated mass ($\frac{d(\Delta m)}{dt}$) is governed by the mass-to-energy conversion efficiency (η):

$$\frac{d(\Delta m)}{dt} = \eta \int_{V_{core}} R_{12} dV \propto \eta \cdot \rho_m^2 \cdot T_c^n \cdot V_{core} \quad (17)$$

Consequently, the total radiative energy emission (Luminosity, L) is derived strictly as a secondary temporal derivative via Einstein's mass-energy equivalence:

$$L = \frac{dE}{dt} = c^2 \frac{d(\Delta m)}{dt} = \eta c^2 \int_{V_{core}} R_{12}(\rho_m, T_c) dV \quad (18)$$

5 Mathematical Summary of the Causal Chain

These rigorous differential derivations establish the following exact causal sequence:

$$\rho_m \xrightarrow{\nabla_\mu T^{\mu\nu}=0} P_c(\rho_m) \xrightarrow{\rho_m \geq \rho_{crit}} \frac{d(\Delta m)}{dt} \xrightarrow{E=\Delta mc^2} L(t) \quad (19)$$

- **Absolute Independent Variable:** Total Mass Density (ρ_m).
- **First-Order Derivative Function:** Sectional Pressure ($P_c \propto \rho_m^{4/3}$).
- **Second-Order Derivative (Byproduct):** Energy Emission Rate ($L = \frac{dE}{dt} \propto f(\rho_m^2)$).